Unit 5, Lesson H1: Definition of Congruence in Terms of Rigid Transformations

Benchmark

MA.912.GR.2.7 Justify the criteria for triangle congruence using the definition of congruence in terms of rigid transformations.

Learning Target

- [ ] I can use the definition of congruence to justify that SSS congruence, SAS congruence, ASA congruence, and AAS congruence are sufficient to show that two triangles are congruent.

5.H1.1 Warm-Up: Geometry as an Animal

1. If geometry were an animal, it would be a …

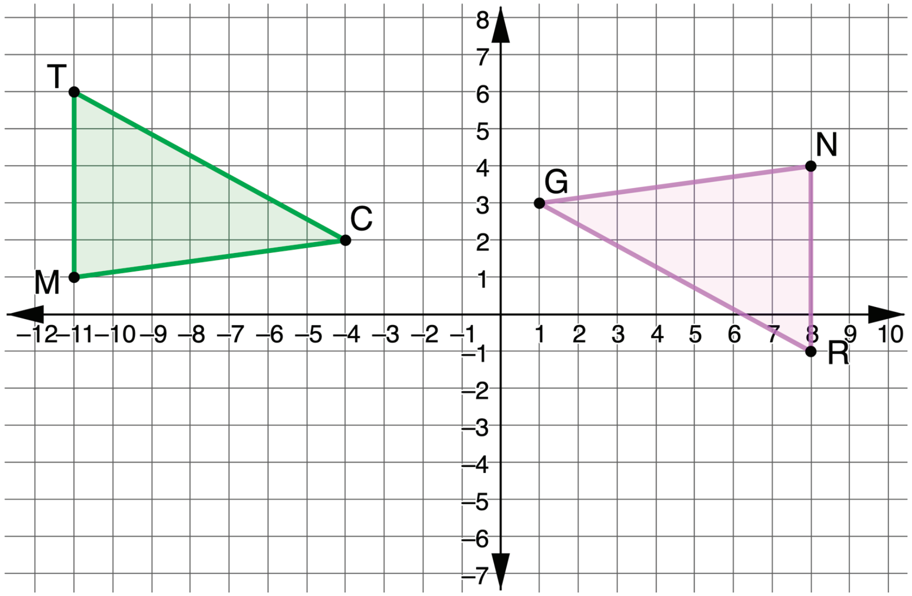
   because …

2. It would live in …

   because …

5.H1.2 Exploration: Justifying Congruence

1. Triangles TCM and RGN are shown, where $\overline{TM} \cong \overline{RN}$, $\overline{MC} \cong \overline{NG}$, and $\angle TMC \cong \angle RNG$.



a. Discuss with your partner what can be concluded from the given information.

b. Ask a classmate for two conclusions they made. Record their conclusions and write your summary of their reason. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.

Conclusion	My Summary of Their Reason	Initials

c. What rigid motions can be applied so that ΔTCM is mapped to ΔRGN?

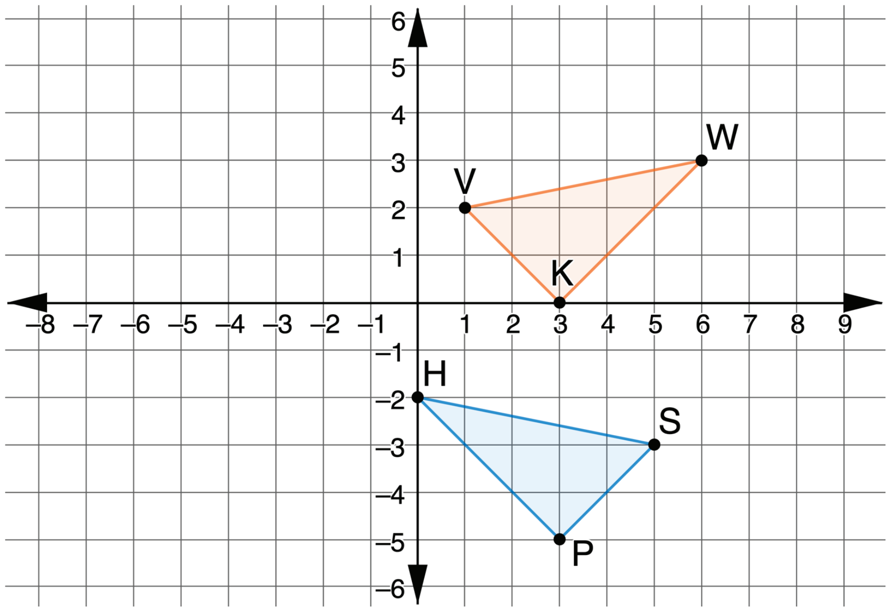
d. Discuss with your partner what can be concluded from the rigid motions.

e. Ask a classmate for two conclusions they made. Record their conclusions and write your summary of their reason. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.

Conclusion	**My Summary of Their Reason**	**Initials**

f. Discuss with your partner why showing that the triangles are congruent using the definition of congruence in terms of rigid transformations shows that SAS congruence is sufficient to prove the triangles are congruent. Summarize your discussion.

2. Triangles VWK and SHP are shown. Assume $\overline{VK} \cong \overline{SP}$, $\overline{WV} \cong \overline{HS}$, and $\overline{KW} \cong \overline{PH}$.



a. Explain how you could use patty paper to verify $\Delta VWK \cong \Delta SHP$.

b. Write a series of transformations that will map ΔVWK onto ΔSHP.

c. Show each on the coordinate grid.
- Draw $\Delta V'W'K'$, the result of the first transformation
- Map the vertices of ΔVWK to $\Delta V'W'K'$
- Map the vertices of $\Delta V'W'K'$ to ΔSHP

d. Complete the statements.

The outcome of mapping the transformations of

ΔVWK to $\Delta V'W'K'$ then from $\Delta V'W'K'$ to Δ______ shows

that V' coincides with
○ H,
○ S,
○ P,
W' coincides

with
○ H,
○ S,
○ P,
and K' coincides with
○ H.
○ S.
○ P.

Because the definition of
○ congruence
○ similarity

in terms of rigid motions preserves distance and

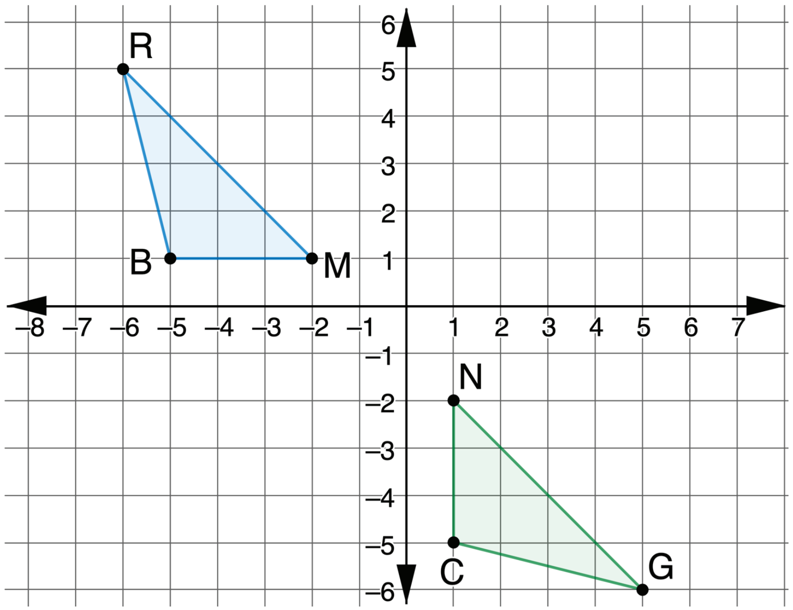
angles, triangle VWK is
○ congruent
○ similar
to triangle

____________. Since $\overline{VK} \cong \overline{SP}$, $\overline{WV} \cong \overline{HS}$, and $\overline{KW} \cong \overline{PH}$

was given,
○ AAS
○ ASA
○ SAS
○ SSS
is sufficient to show triangles

are congruent.

3. Triangles RMB and GNC are shown, where $\overline{RM} \cong \overline{GN}$, $\angle BRM \cong \angle CGN$, and $\angle BMR \cong \angle CNG$.



Adriana knows that there is a series of rigid motions that maps ΔRMB onto ΔGNC.

a. Select all true statements after the transformations.

☐ Line segment RB coincides with line segment GN.
☐ Line segment BM coincides with line segment CN.
☐ Line segment CG coincides with line segment BR.
☐ Angle NCG coincides with angle RMB.
☐ Angle RBM coincides with angle GCN.

b. Explain how the given information and the definition of congruence in terms of rigid motions shows that $\Delta RMB \cong \Delta GNC$.

5.H1.3 Wrap-Up: Reflection

1. What is something that is now "squared away" in your mind about the definition of congruence in terms of rigid transformations?

2. What are three "points" that are important for showing triangle congruence using the definition of congruence in terms of rigid transformations?

3. What is something that is still "circling" your brain about the definition of congruence in terms of rigid transformations?